EE322: Analog and Mixed Signal Circuits

LTspice Simulation Report

Low-Dropout CC-CV 12 V Lead-Acid Battery Charger with Auto Trickle Charge

Item	Details
Team name	Team Brownian
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Project	All-analog CC-CV battery charger with trickle charge and status LEDs

1. Circuit Schematic

Figure 1 shows the complete LTspice schematic of the charger. The circuit is a fully analog, low-dropout CC-CV (constant current, constant voltage) charger for a 12 V lead-acid battery. A P-channel MOSFET (M1, AO6407) is the pass element, and two LM2901 comparators drive the status LEDs.

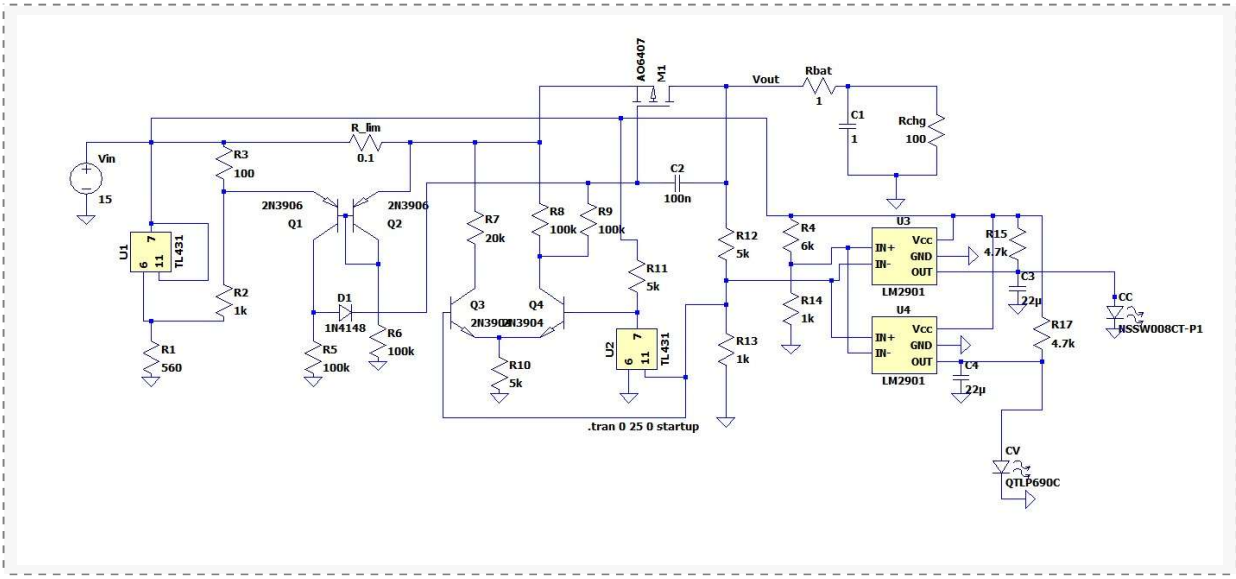


Figure 1: Complete LTspice schematic of the CC-CV charger with comparator-based LED indication

1.2 Simulation Conditions

Parameter	Setting
Input supply	Vin = 15 V DC
Analysis	.tran 0 25 0 startup (0 to 25 s, startup option enabled)

Parameter	Setting
Battery model	Rbat = 1 Ω in series with C1 = 1 F, with Rchg = 100 Ω in parallel as the charge-dependent load, referenced to ground
Comparator supply	LM2901 Vcc taken from the 15 V input rail
Input signal type	DC supply only (no AC or pulsed input). The dynamic behaviour comes from the battery model charging up.

2. Working Principle and Circuit Explanation

2.1 Purpose

The circuit charges a 12 V lead-acid battery without a microcontroller. It limits the current while the battery is deeply discharged (constant current), holds the voltage at a safe ceiling as the battery fills (constant voltage), and drops to a small maintenance current once the battery is full (trickle). Because the pass element is a P-channel MOSFET, the dropout voltage is only the MOSFET drop plus the sense-resistor drop, so the charger keeps regulating even when Vin is not much above the battery voltage.

2.2 Major Blocks

- Reference and constant-current block (U1, Q1, Q2, R1, R2, R3, D1): The TL431 and resistors set a reference voltage. Q1 and Q2 monitor the voltage across \(R_{lim}\) and reduce the MOSFET drive when the charging current reaches the set limit. D1 allows either the current or voltage control loop to take control.

- Constant-voltage block (Q3, Q4, U2, R7–R11): Q3 and Q4 compare the battery voltage with a 2.5 V reference from the TL431. When the battery reaches the set voltage, the circuit reduces the charging current to keep the voltage constant.

- Pass element and output stage (M1, C2): The AO6407 MOSFET controls the charging current. C2 helps keep the circuit stable.

- Battery model and sensing R_lim, R_bat, C1, R_chg): These components represent the battery in simulation. R_lim measures the charging current, and the R12/R13 divider measures the battery voltage.

- CC/CV indicator block (U3, U4, R4, R14, R15, R17, C3, C4): The LM2901 comparators control the CC and CV LEDs based on the battery voltage. The capacitors help prevent the LEDs from flickering when the charging mode changes.

2.3 Design Change from the Proposal

In the proposal, the charge and trickle LEDs were switched by 2N7000 N-channel MOSFETs with a zener (1N754) and 1N4148 diodes. In this simulation those MOSFET stages have been replaced by two LM2901 comparators (U3, U4). Comparators give a sharp, repeatable threshold, remove the dependence on MOSFET gate thresholds, and make the LED state change cleanly for the three stages. The power stage and the CC/CV control loops are unchanged in principle.

3. Components Used

All components in the schematic are listed below. "Ideal" parts are the basic LTspice resistor, capacitor and DC source. Active parts use SPICE models. Items marked "Simulation only" are part of the battery/load model and are not built in hardware.

Component	Value / Rating	Model / Type	Source
Rbat	1 Ω	Ideal resistor	EE Lab
C1	1 F	Ideal capacitor	EE Lab
Rchg	100 Ω	Ideal resistor	EE Lab
R_lim	0.1 Ω	Ideal resistor	EE Lab
R1	560 Ω	Ideal resistor	EE Lab
R2, R13, R14	1 kΩ	Ideal resistor	EE Lab
R3	100 Ω	Ideal resistor	EE Lab
R4	6 kΩ	Ideal resistor	EE Lab
R5, R6, R8, R9	100 kΩ	Ideal resistor	EE Lab
R7	20 kΩ	Ideal resistor	EE Lab
R10, R11, R12	5 kΩ	Ideal resistor	EE Lab
R15, R17	4.7 kΩ	Ideal resistor	EE Lab
C2	100 nF	Ideal capacitor	EE Lab
C3, C4	22 µF	Ideal capacitor	EE Lab
M1	AO6407	P-channel MOSFET, manufacturer SPICE model	Robu
Q1, Q2	2N3906	PNP BJT, LTspice library model	Robu
Q3, Q4	2N3904	NPN BJT, LTspice library model	Robu
D1	1N4148	Switching diode, LTspice library model	Robu
U1, U2	TL431	Programmable shunt reference, SPICE model	Robu
U3, U4	LM2901	Quad comparator, manufacturer SPICE model	Using LM339 from EE Lab
CC and CV LED		LTSpice library model	LED from EE Lab

4. Simulation Results and Plots

The transient simulation runs from 0 to 25 s with Vin = 15 V. The plots below show the charger moving through its three stages.

4.1 Battery Voltage and Charging Current

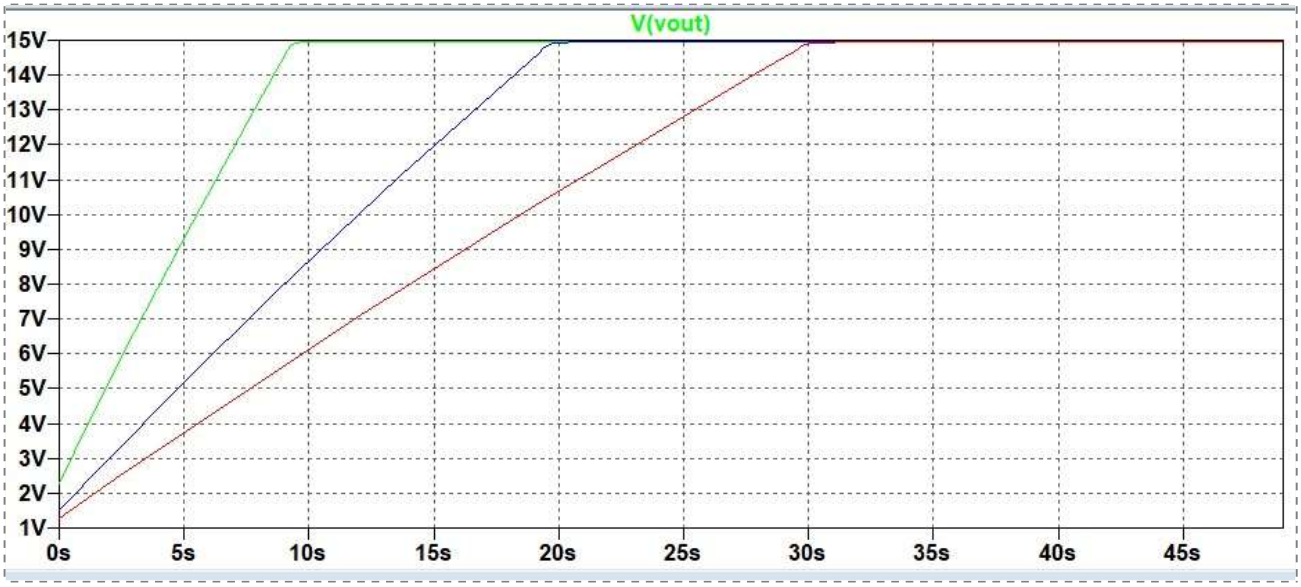


Figure 2: Battery Voltage at various capacitances (0.5 1 1.5)

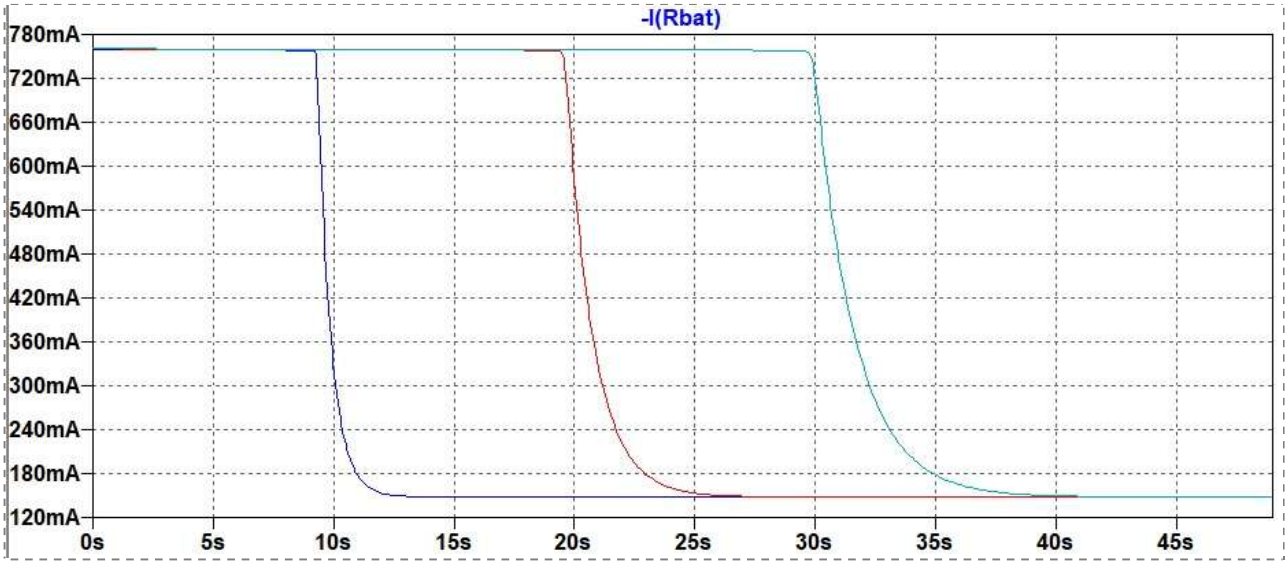


Figure 3: Current through R_bat at various capacitances (0.5 1 1.5)

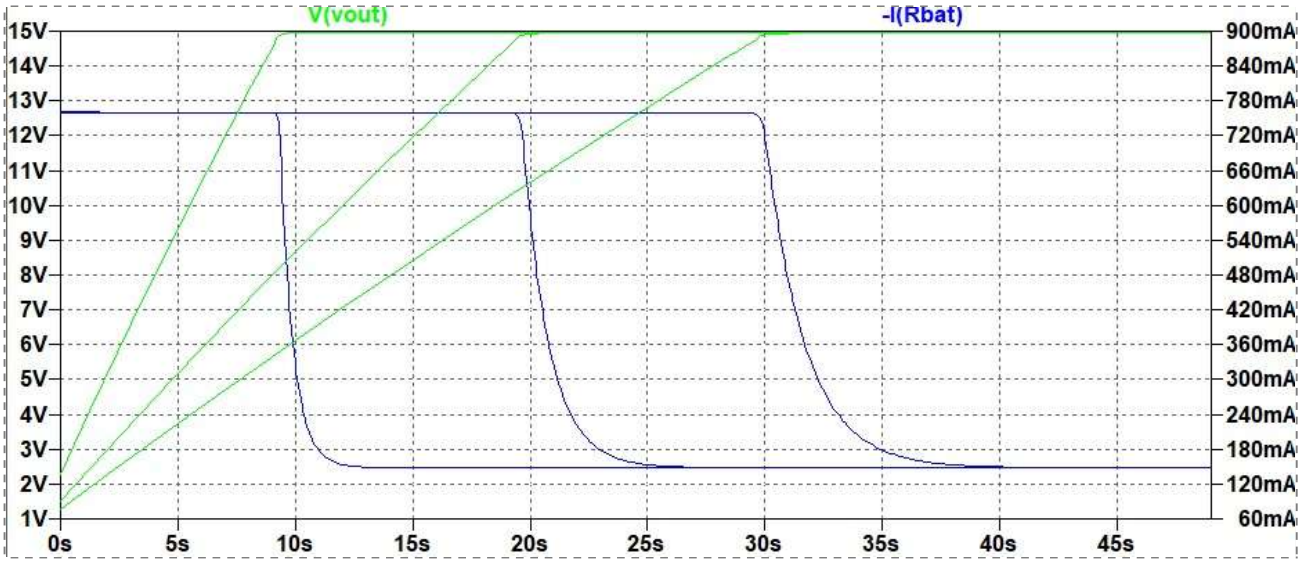


Figure 4: CC CV trickle stages at various capacitances (0.5 1 1.5)

5. Discussion and Conclusion

5.1 Summary of Results

We can see in the simulation that the initially the current remains constant and the voltage riss till supply voltage and then the current starts decreasing and then it also becomes constant. So it goes from constant current to constant voltage and finally the trickle stage where the battery voltage is maintained at 15V with very small amount of current.